Design Challenge - 2

Aim - The purpose of the design challenge is to understand the geometry and design of the replicated mandibular plate by using cardboard, aluminium sheet, brass fasteners and other materials. Further, the task was to redesign the baseline fixation plate, define the failure criteria and test it to prove that the redesign plate was better.

Methods

1. A functional unit design is a design where a simple fracture is corrected by placing a plate spanning across both fragments of the fracture and securely fastening it using a brass fastener on each fragment. The strut region between the holes is where most of the torsional stress is concentrated when bent [1]. As [2] states, holes are "*stress risers*" which means that they amplify stress. This is because they cause geometric discontinuity, distributing stress to neighboring cross-sectional areas. When distance between two holes is increased, the stress is spread more uniformly across the strut region, increasing stiffness and rigidity of the strut region[2]. Hence, instead of having two holes 0.5 cm away in each direction from the vertical midline, we chose to have two holes 1 cm away from the vertical midline in each direction.



Figure 1 *- Redesigned Plate*

2. & 3.

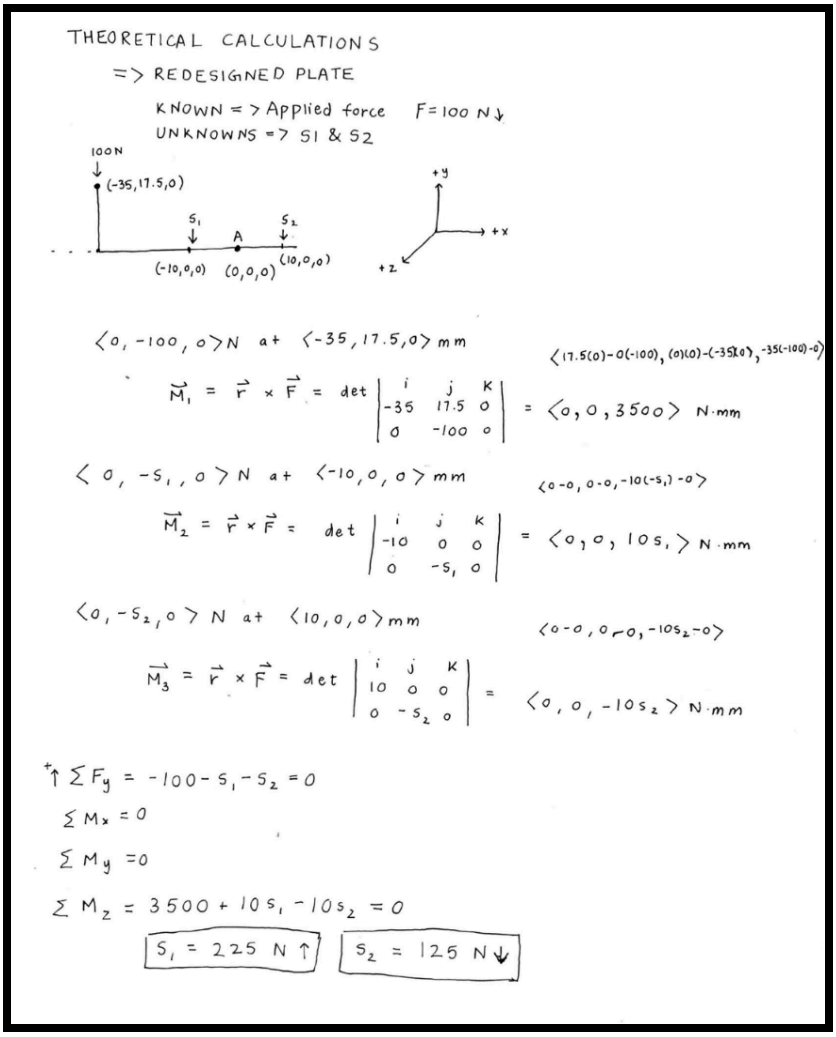


Figure 2 *- FBD of theoretical model and calculations for S1 and S2*

4. We made modifications to Section 9 of the ASTM standard. For section 9, instead of having five test samples, we had two samples (one was baseline sample and one was redesigned sample) tested three times to get consistent and accurate data. Other conditions like labelled test plates, new and untested samples etc. were followed.

5. To elaborate on the previous question, the load was applied in the negative Y-direction, which was directly clamped to the corner of the second fragments (X2 and B2). First, the mass hook was attached using a paper clip (its mass was included which was 5g). Next, mass blocks of 10grams were added until failure was noticed. Failure was defined as an angular displacement of 10 degrees or wider between the two fragments.

The reason this failure criteria was suitable for the redesigned plate was because angular displacement for a specific angle was more evident than linear displacement. Additionally, when weights were loaded on the plate, the fracture widened non-uniformly and hence angular

displacement was a more accurate measurement. Had the widening been more uniform, then displacement would have been a better measurement. To draw comparison with the real-life scenario, when there is an angular displacement to the lower jawbone, it can cause wearing of teeth, crossbites and other such chewing inefficiencies, pain and sensitivity [3].

6. According to [4], an angular displacement of lesser or equal to 1 degree can cause severe impact to molars. However, since an angular displacement of a 1 degree is difficult to notice with a protractor, we chose a larger angle so that it was easier to notice. Therefore, our failure criteria was an angular displacement of 10 degrees or wider which is also clinically

7 and 8.

Assumption - The team already has made the baseline design model and has some previous knowledge from it about the tools and method. Mass of hook is 5 grams

Phase 1) Building the redesigned plate

a) Take two fragments of the “fracture” (B1 and B2) and align them side-by-side such that B1 is on the left and B2 on the right
b) Next, take the aluminum flashing material and cut a rectangular piece of 150mm X 30mm and then fold it by half by the length so that the width is 15mm now and length remains the same
c) Fold the rectangle again into half so that the length is 75mm and width is same
d) Cut the rectangle into half by its width so that you have two rectangles each with length as 37.5mm and width as 15mm.
e) Mark the horizontal and vertical midlines
f) At the intersection of B1 make a marking where the hole will be made using te the hole-punch
g) Repeat steps (f) for fragment B2
h) Bring both fragments close and align the first aluminium piece by its vertical over both the fragments
i) Mark a circle on the fragments where the hole should be made and make a tiny hole using the brass fasteners
j) Secure the aluminium rectangle with the brass fastener
k) Repeat the same procedure for the flipped side

Phase 2) Testing the plates

a) To test the baseline design first, clamp it on the mini-bench vise and tighten it by lowering the suction lever
b) Next, place the hook on which the masses will be loaded using a paperclip on the second fragment (B2) by widening the paper clip
c) Ensure no initial gap between the fragments for accurate measurement
d) Start by adding a 10g mass and measure the angular displacement after its addition
e) Keep adding masses until the angular displacement is 10 degrees or more

f) Record the mass value + mass of hook (5g) when “failure” is achieved
g) Repeat step (d) to (f) two more times and record data each time
h) Repeat steps (a) through (g) for the redesigned model for a total of three trials.

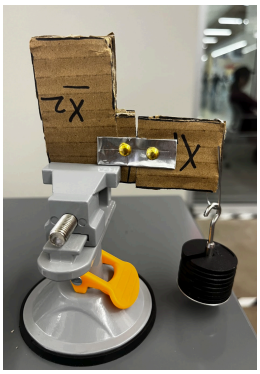


Figure 3 *- Test setup image*

Challenges -

a) The paperclip on which the mass hook was hooked kept slipping. Hence, we had to widen the paper clip enough such that the width of the paper clip was as much as the thickness of the cardboard. Had the paperclip been too narrow, it would not fit in and if it had been too thick, it would slip.
b) Securely inserting the brass fastener was a challenge because the gloves were too thick. Hence we had to insert them by twisting it.

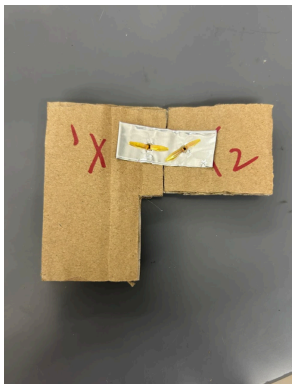
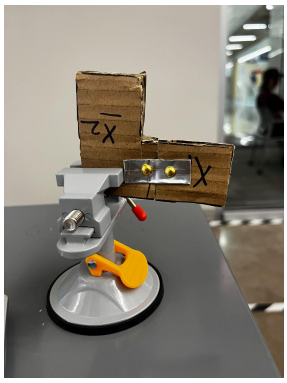
Safety considerations -

a) Always wear scratch resistant gloves, especially when cutting aluminum sheets and inserting brass fasteners.

Changes -

No changes made

Results

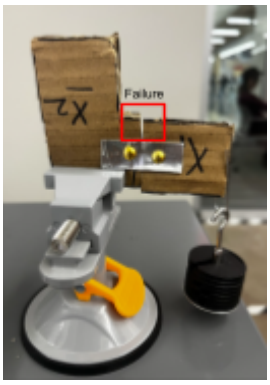


Figure 4 to 6 (left to right) *- baseline plate, baseline plate before experiment, baseline plate after failure*

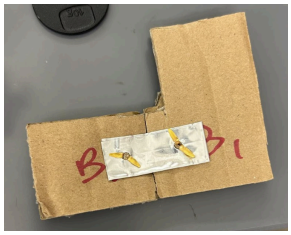



Figure 7 & 8 (left to right) *- redesigned plate, redesigned plate after failure*

2.

Table 1 *- Baseline plate vs Redesigned plate experiment data*

	Baseline Design Model	**Redesigned Model**
Trial 1 Mass (in grams)	45	105
Trial 2 Mass (in grams)	35	95
Trial 3 Mass (in grams)	55	135
Avg Mass (in grams)	45	111.67
Stdev Mass (in grams)	10	20.82

3.

Table 2 *- Baseline vs Redesigned plate comparison of theoretical and experimental data*

	Baseline Plate	**Redesigned Plate**
Theoretical S1 for 100N force (in N)	400	225
Theoretical S2 for 100N force (in N)	300	125
Experimental Failure Average Mass (in g)	45	111.67

For the baseline plate, the S1 and S2 forces were high, translating to lower force distribution to the strut region. By lowering the S1 and S2 forces by increasing the distance from the midline, greater force distribution on the strut region is ensured. Higher force distribution on the strut region leads to increased mass before achieving “failure”.

Discussion

1. Based on Table 2, theoretically, S1 and S2 should decrease, forces in the strut region should increase and the plate with lower S1 & S2 force should be able to bear greater mass before achieving “failure”. This is because when distance of the stress risers increases from the midline, the lever arm for the load to distribute increases. This leads to better load distribution, increasing the stiffness of the strut region. This makes the strut

more resistant to mass, increasing the average mass it can withstand before an angular displacement of 10 degrees.

2. We predicted that S1 and S2 should decrease for the redesigned plate and it did as evident from Table 1. The redesigned plate had an angular displacement of 10 degrees when 111.67g of mass was loaded as compared to 45g of mass for the baseline model. This is a positive result because the redesigned plate can withstand more weight than "failing".

   The reason for the result is because the baseline design of the screw holes were 0.5 cm apart horizontally from the vertical midline, causing a less uniform stress distribution. This lowered the plate stiffness compelling the baseline model to "fail" with an average of only 45g of mass. This finding is reinforced by [3] that reported that stress concentration around the hole was highest when it was near the fracture line of the fragments and decreased as distance from the fracture line increased due to uniform stress distribution.

3. Answered in question 2

4. From my results, I understand that the positioning of screws and holes is not arbitrary. There is a lot of ground work in selecting the right position of the screw. For instance, making a 0.5cm difference in my screw position can make the plate resilient to higher mass. This can potentially affect the life of so many patients as the baseline for failure is high. Further, the smaller the hole, the better it is because larger holes are larger stress risers and cause greater force distribution as a larger area is poked out. In a clinical setting, larger holes on the plate can cause high screw forces S1 and S2 which can potentially harm the patient. Similarly, the length and tightness of the brass fastener matters as well. Had the brass fastener been shorter and looser, then the fragments would have been split apart and in a clinical setting it could lead to serious trouble and potential surgery.

5. The two potential sources of errors are -
   a. **Inconsistent weight application** - The weight (due to the 10g masses) was placed on a rod that was hooked to a paper clip, clipping the cardboard piece. Since the paper clip was only clipped without being glued, it was loose and would move every time the rod got heavier. This resulted in a slight difference in weight application which could potentially affect the data. If the clip moved outward (in the positive X direction), lesser weight would be noticed while movement in the negative X direction would lead to higher weight as guided by moment.
   b. Angle measurement- To measure angular displacement we used a protractor. The zero of the protractor would be right above the midline. Since there were no clamps for the protractor, it was held by a teammate. Due to human error the 0 degree mark could potentially not have always been at the same position as it should. This may cause some measurement error.

6. This design challenge posed several learning opportunities for me. To start with, I could directly connect the 3D Reaction Based Equilibrium (RBE) concept to a real-life biomedical application. I got to fine-tune my mathematical skills by using practical numbers from human physiology. Next, I learnt several important design concepts. In BMEN 207, we were taught about different types of holes and how useful they could be when designing. But during this challenge, I realised how holes are “stress risers” and increase force in the neighboring areas. We were also taught about different types of screws like M5,M6 etc. but were never taught their application in a medical environment. Hence, I could relate the use of screws to a clinical concept. For my internship, I used to perform testing of catheters and would frequently wonder why the failure criteria the way it is. And now, through this design challenge I had the opportunity to understand how to build a test, define failure criteria and reason it.

References -

[1]ASTM International, *ASTM F3437-23: Standard Test Methods for Metallic Bone Plates Used in Small Bone Fracture Fixation*, West Conshohocken, PA, USA: ASTM International, 2023, doi: 10.1520/F3437-23
[2] J. Yoo, “Research update on stress riser fractures,” *Indian Journal of Orthopaedics*, vol. 54, no. 2, pp. 123–131, 2020, doi: 10.1007/s43465-020-00046-2.
[3] T. Kimura, T. Sakoda, K. Ogura, and Y. Iida, “A comparative study of the most suitable miniplate fixation for mandibular symphysis fracture using a finite element model,” *Journal of Oral and Maxillofacial Surgery*, vol. 64, no. 10, pp. 1536–1540, Oct. 2006, doi: 10.1016/j.joms.2006.05.043.
[4] Y. Tang and R. Yarong, “Effect of Angular Malalignment on Occlusal Displacement in Mandibular Fractures,” *Technical Report*, University of Georgia, Athens, GA, USA, 2018. [Online]. Available: https://www.ai.uga.edu/sites/default/files/inline-files/tang_yarong.pdf